

Matrix Spencer for Toeplitz, Hankel, and Finite-Entropy Lift Classes

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Abstract

The Matrix Spencer conjecture asks whether, for every collection of self-adjoint $n \times n$ matrices $A_1, \dots, A_n$ with $\|A_i\|_{\text{op}} \leq 1$, one can choose signs $\varepsilon_i \in \{-1, 1\}$ so that $\|\sum_i \varepsilon_i A_i\|_{\text{op}} = O(\sqrt{n})$. We prove Matrix Spencer bounds for Toeplitz, Hankel, and related finite-entropy structured classes. The main Toeplitz consequence is the full scalar theorem: every collection of arbitrary complex $d \times d$ Toeplitz contractions satisfies

$$\text{disc}_{\text{op}}(A_1, \dots, A_n) \leq C \sqrt{n \log \left(e + \frac{d}{n} \right)}.$$

The same bound holds for arbitrary complex Hankel contractions. Hence self-adjoint scalar Toeplitz and Hankel matrices satisfy the conjectured $O(\sqrt{n})$ bound throughout the square regime $d \leq n$. The proof uses the finite Toeplitz extension theorem, de la Vallée Poussin localization, and a sampled multiplier lift to a diagonal algebra of size $O(d)$.

We also isolate the general mechanism behind the proof. If a tuple admits a K -bounded lift to a finite block algebra $\bigoplus_{\alpha=1}^m M_{r_\alpha}(\mathbb{C})$, with $r_* = \max_\alpha r_\alpha$ and $R = \sum_\alpha r_\alpha$, then

$$\text{disc}_{\text{op}}(A_1, \dots, A_n) \leq CK \sqrt{n \log \left(e + \frac{r_* R}{n} \right)}.$$

This finite-entropy lift theorem yields fixed-block and fixed-level block Toeplitz/Hankel bounds, common finite-dimensional C^* -algebra bounds, finite-group convolution bounds with controlled irreducible dimensions, and stability under compressions, quotients, bounded sums, and fixed-size tensor amplification. We retain the banded, tame-symbol, and scale-localized top-band results as finite-section refinements: banded Toeplitz matrices of bandwidth w satisfy $C\sqrt{n \log(e + w/n)}$ whenever the section resolves the bandwidth, and full-bandwidth symbols with derivative $O(d)$ satisfy $O(\sqrt{n})$ for $d \leq n$.

Keywords. Matrix discrepancy; Matrix Spencer conjecture; Toeplitz matrices; Hankel matrices; finite-dimensional C^* -algebras; finite group convolution; finite-entropy lifts; trigonometric polynomials; finite sections; vector discrepancy; Schur multipliers.

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1 Introduction

Spencer’s theorem that “six standard deviations suffice” is one of the central results of classical discrepancy theory [25]. In one standard form, for n vectors in the unit cube of $\mathbb{R}^n$, signs can be chosen so that the resulting signed sum has ℓ_∞ norm $O(\sqrt{n})$. Rectangular extensions give the bound $O(\sqrt{n \log(e + m/n)})$ for n vectors in ℓ_∞^m ; see, for example, [10, 23] and the independent work of Gluskin [14]. The Matrix Spencer conjecture asks for the analogous assertion with the ℓ_∞ norm replaced by the spectral norm and vectors replaced by self-adjoint matrices.

Definition 1.1 (Matrix discrepancy). For self-adjoint matrices $A_1, \dots, A_n$, set

$$\text{disc}(A_1, \dots, A_n) = \min_{\varepsilon \in \{-1, 1\}^n} \left\| \sum_{i=1}^n \varepsilon_i A_i \right\|_{\text{op}}.$$

Question 1.2 (Matrix Spencer). *Is there a universal constant C such that, for all n and all self-adjoint $n \times n$ matrices $A_1, \dots, A_n$ satisfying $\|A_i\|_{\text{op}} \leq 1$,*

$$\text{disc}(A_1, \dots, A_n) \leq C\sqrt{n}?$$

More generally, for $d \times d$ matrices with $d \leq n$, one asks for the same bound with a constant independent of d . When d is much larger than n , a logarithmic dependence on d/n is unavoidable already for diagonal matrices, by the rectangular vector discrepancy problem.

The general conjecture remains open. Classical noncommutative Khintchine inequalities imply random-sign bounds with logarithmic dimensional losses [21]. The intrinsic-freeness matrix concentration inequality of Bandeira, Boedihardjo and van Handel [3] has led to major progress on matrix discrepancy. Hopkins, Raghavendra and Shetty proved an $O(\sqrt{n})$ bound under a Frobenius-norm hypothesis and connected the conjecture to quantum communication [17].

Bansal, Jiang and Meka proved the conjectured $O(\sqrt{n})$ bound for matrices of rank at most $n/\log^3 n$ [8]. Dadush, Jiang and Reis developed a mirror-descent framework giving strong bounds for low-rank and block-diagonal matrices and for Schatten-norm variants [13]. Subsequent intrinsic-freeness estimates sharpen the general concentration input used in this area [4]. Very recent work also studies structured group-representation and finite-dimensional C^* -algebra instances of Matrix Spencer [5, 1]. These results still leave many full-rank, highly structured finite-dimensional families outside the known theory. The results below show that scalar Toeplitz and Hankel matrices, and more generally finite-entropy lift classes, form a large tractable region beyond the low-rank/Frobenius route.

This paper treats finite Toeplitz sections. A real-valued trigonometric polynomial

$$p(\theta) = \sum_{|k| < d} c_k e^{ik\theta}, \quad c_{-k} = \overline{c_k},$$

defines the $d \times d$ self-adjoint Toeplitz section

$$T_d(p) = (c_{x-y})_{1 \leq x, y \leq d}.$$

The real symmetric case is the special case $c_k \in \mathbb{R}$, equivalently p is even. If $c_k = 0$ for $|k| > w$, we say the bandwidth is at most w . Such matrices can have rank d and Frobenius norm squared of order d even when $\|T_d(p)\|_{\text{op}} \leq 1$; hence, when $d \gg n/\log^3 n$, the available low-rank hypotheses do not apply. For background on Toeplitz matrices and their symbols see, for example, [16, 9, 15].

Results

The strongest result is the full scalar Toeplitz theorem. It is most naturally stated for arbitrary complex matrices, using disc_{op} .

Theorem 1.3 (Full scalar Toeplitz and Hankel Matrix Spencer). *There is a universal constant C such that the following holds. Let $A_1, \dots, A_n \in M_d(\mathbb{C})$ be arbitrary complex Toeplitz matrices with $\|A_i\|_{\text{op}} \leq 1$. Then*

$$\text{disc}_{\text{op}}(A_1, \dots, A_n) \leq C \sqrt{n \log \left(e + \frac{d}{n} \right)}.$$

The same estimate holds for arbitrary complex Hankel matrices. In particular, self-adjoint scalar Toeplitz and Hankel matrices satisfy Matrix Spencer whenever $d \leq n$.

The proof appears in Section 6. It uses a bounded L^∞ inducer for each finite Toeplitz contraction, smooths the inducer to a degree- $O(d)$ trigonometric polynomial without changing the relevant Toeplitz coefficients, and then signs the values on an $O(d)$ -point norming grid. This is a bounded finite lift to a diagonal algebra of entropy $O(d)$.

More generally, the Toeplitz proof is an instance of a finite-entropy lift principle.

Theorem 1.4 (Finite block-lift Matrix Spencer, informal form). *Let $\mathcal{B} = \bigoplus_{\alpha=1}^m M_{r_\alpha}(\mathbb{C})$, let $r_* := \max_\alpha r_\alpha$, $R := \sum_\alpha r_\alpha$, and suppose $A_1, \dots, A_n$ admit K -bounded lifts to $\mathcal{B}$ through a contraction on the span of the lifts. Then*

$$\text{disc}_{\text{op}}(A_1, \dots, A_n) \leq CK \sqrt{n \log \left(e + \frac{r_* R}{n} \right)}.$$

If the matrices and lifts are self-adjoint, the same statement holds for disc .

Consequences include fixed-block and fixed-level block Toeplitz/Hankel matrices, common finite-dimensional C^* -algebras with small reduced entropy, finite abelian convolution, bounded irreducible-dimension group convolution, Toeplitz-plus-Hankel classes with bounded decomposition, and pathwise normalized random samples from these structured classes.

Our banded theorem is a finite-section refinement of the full Toeplitz theorem. It proves the optimal entropy w directly from the finite section, without using a global Toeplitz extension.

Our first theorem proves the conjectured Matrix Spencer bound for low-frequency Toeplitz symbols whenever the finite section resolves the bandwidth by a fixed absolute factor.

Theorem 1.5 (Banded Toeplitz Matrix Spencer). *There is a universal constant C such that the following holds. Let $w \geq 1$ and $d \geq 20w$. Let*

$$A_i = T_d(p_i), \quad i = 1, \dots, n,$$

be self-adjoint $d \times d$ Toeplitz matrices of bandwidth at most w , and assume $\|A_i\|_{\text{op}} \leq 1$ for every i . Then there exist signs $\varepsilon \in \{-1, 1\}^n$ such that

$$\left\| \sum_{i=1}^n \varepsilon_i A_i \right\|_{\text{op}} \leq C \sqrt{n \log \left(e + \frac{w}{n} \right)}.$$

Consequently, if $w \leq n$, then

$$\text{disc}(A_1, \dots, A_n) \leq C \sqrt{n}.$$

The proof is deliberately short. The main point is not a new partial-coloring algorithm but a norm model. In the range $d \geq 20w$, the Toeplitz section sees the sup norm of its symbol:

$$\frac{1}{2} \|p\|_{L^\infty(\mathbb{T})} \leq \|T_d(p)\|_{\text{op}} \leq \|p\|_{L^\infty(\mathbb{T})}.$$

Thus operator-norm normalization of $T_d(p_i)$ implies uniform boundedness of the symbols p_i . Sampling the symbols on a grid of $8w$ points reduces the problem to signing bounded vectors in ℓ_∞^{8w} , where vector Spencer applies. A Bernstein inequality for trigonometric polynomials transfers the grid bound back to $L^\infty(\mathbb{T})$, and the elementary Toeplitz upper bound transfers the symbol estimate back to matrices.

The same finite-section idea gives a second theorem that is closer to the full-bandwidth endpoint. It replaces bandlimitedness by a scale-invariant smoothness assumption.

Theorem 1.6 (Tame full-bandwidth Toeplitz Spencer). *There is a universal constant C such that the following holds. Let $A_i = T_d(p_i)$ be self-adjoint $d \times d$ Toeplitz matrices with $\deg p_i < d$ and $\|A_i\|_{\text{op}} \leq 1$. Suppose*

$$\max_{1 \leq i \leq n} \|p'_i\|_\infty \leq L.$$

Then

$$\text{disc}(A_1, \dots, A_n) \leq C \left(1 + \frac{L}{d} \right) \sqrt{n \log \left(e + \frac{d}{n} \right)}.$$

In particular, if $d \leq n$ and $L = O(d)$, then

$$\text{disc}(A_1, \dots, A_n) = O(\sqrt{n}).$$

Theorem 1.6 includes symbols of full algebraic degree. What matters is not the largest Fourier mode by itself but whether the symbol can spike on angular scales smaller than the finite section can test.

We also record two full-bandwidth reductions. The first is the direct symbol-normalized version.

Proposition 1.7 (Symbol-normalized full bandwidth). *There is a universal constant C such that, if $A_i = T_d(p_i)$ are self-adjoint Toeplitz $d \times d$ matrices of arbitrary bandwidth and $\|p_i\|_{L^\infty(\mathbb{T})} \leq K$, then*

$$\text{disc}(A_1, \dots, A_n) \leq CK \sqrt{n \log \left(e + \frac{d}{n} \right)}.$$

In particular, for $K = O(1)$ and $d \leq n$, this gives $\text{disc}(A_1, \dots, A_n) = O(\sqrt{n})$.

The next result records a second reduction for the full-bandwidth endpoint. It shows that the symbol-recovery problem is not spread across all frequencies: after low-pass localization, every possible loss lives in a bounded number of top dyadic bands.

Proposition 1.8 (Scale-localized recovery and top-band reduction). *There is a universal constant C such that the following holds. Let $d \geq 504$ and let p be a real-valued trigonometric polynomial of degree $< d$. After de la Vallée Poussin localization, every low-frequency piece at scale at most $d/504$ obeys*

$$\|L_m p\|_\infty \leq 6 \|T_d(p)\|_{\text{op}} \quad (504m \leq d),$$

and every dyadic piece obeys

$$\|\Pi_j p\|_\infty \leq 12 \|T_d(p)\|_{\text{op}} \quad (504 \cdot 2^j \leq d).$$

Define

$$\kappa(d) = \sup\{\|p\|_\infty : p \text{ real-valued, } \deg p < d, \|T_d(p)\|_{\text{op}} \leq 1\}$$

and let $\kappa_{\text{band}}(d)$ be the same supremum restricted to one dyadic band contained in $[d/4032, 2d]$. Then

$$\kappa(d) \leq 6 + C \kappa_{\text{band}}(d).$$

Consequently, for arbitrary self-adjoint Toeplitz matrices $A_i = T_d(p_i)$ with $\|A_i\|_{\text{op}} \leq 1$,

$$\text{disc}(A_1, \dots, A_n) \leq C(1 + \kappa_{\text{band}}(d)) \sqrt{n \log \left(e + \frac{d}{n} \right)}.$$

In particular, a dimension-free top-band recovery bound $\kappa_{\text{band}}(d) = O(1)$ would imply Toeplitz Spencer at the optimal $C\sqrt{n}$ scale whenever $d \leq n$.

2 Preliminaries

We write $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ and use normalized Haar measure $d\theta/(2\pi)$ on $\mathbb{T}$. For a trigonometric polynomial p , write

$$\|p\|_\infty = \sup_{\theta \in \mathbb{T}} |p(\theta)|.$$

All matrices in the main statements are self-adjoint. The real symmetric case follows by restricting to real even symbols.

2.1 The diagonal signing input

The only discrepancy-theoretic black box used in the banded and tame proofs is the following standard rectangular extension of Spencer's vector discrepancy theorem.

Theorem 2.1 (Vector Spencer, diagonal form). *There is a universal constant C_S such that, for all $x_1, \dots, x_n \in \mathbb{C}^m$ with $\|x_i\|_\infty \leq 1$, there are signs $\varepsilon \in \{-1, 1\}^n$ satisfying*

$$\left\| \sum_{i=1}^n \varepsilon_i x_i \right\|_\infty \leq C_S \sqrt{n \log \left(e + \frac{m}{n} \right)}.$$

Equivalently, if D_i are diagonal $m \times m$ matrices with $\|D_i\|_{\text{op}} \leq 1$, then

$$\text{disc}(D_1, \dots, D_n) \leq C_S \sqrt{n \log \left(e + \frac{m}{n} \right)}.$$

For complex vectors, apply the real result to real and imaginary parts and absorb the resulting absolute constant. The theorem follows from the usual form of Spencer's six-standard-deviations theorem and its rectangular extensions [25, 10, 23]; algorithmic versions are available through the work of Bansal [7], Lovett and Meka [20], and subsequent refinements such as [19].

2.2 Toeplitz sections and the easy symbol bound

Let

$$p(\theta) = \sum_{|k| < d} c_k e^{ik\theta}, \quad c_{-k} = \overline{c_k}.$$

The $d \times d$ Toeplitz section of p is

$$T_d(p) = (c_{x-y})_{1 \leq x, y \leq d}.$$

The next lemma is standard; we include the proof to keep the manuscript self-contained.

Lemma 2.2 (Toeplitz upper bound). *For every $f \in L^\infty(\mathbb{T})$ and every $d \geq 1$, the finite Toeplitz section $T_d(f)$ with entries $\widehat{f}(x - y)$ satisfies*

$$\|T_d(f)\|_{\text{op}} \leq \|f\|_\infty.$$

If f is real-valued, then $T_d(f)$ is self-adjoint.

Proof. For $z = (z_1, \dots, z_d)$ and $y = (y_1, \dots, y_d)$ in $\mathbb{C}^d$, set $q_z(\theta) = \sum_{x=1}^d z_x e^{ix\theta}$ and define q_y similarly. Since $\widehat{f}(k)$ are the Fourier coefficients of f ,

$$\langle T_d(f)z, y \rangle = \int_{\mathbb{T}} f(\theta) q_z(\theta) \overline{q_y(\theta)} \frac{d\theta}{2\pi}.$$

Therefore, by Cauchy–Schwarz and Parseval,

$$|\langle T_d(f)z, y \rangle| \leq \|f\|_\infty \|q_z\|_{L^2(\mathbb{T})} \|q_y\|_{L^2(\mathbb{T})} = \|f\|_\infty \|z\|_2 \|y\|_2.$$

Taking the supremum over unit z, y gives the operator-norm bound. If f is real-valued, then $\widehat{f}(-k) = \overline{\widehat{f}(k)}$, so the Toeplitz section is self-adjoint. $\square$

We shall also use Bernstein's inequality for trigonometric polynomials; see, for example, Zygmund [28, Chapter X].

Lemma 2.3 (Bernstein inequality). *If P is a trigonometric polynomial of degree at most r , then*

$$\|P'\|_\infty \leq r \|P\|_\infty.$$

Consequently, for $\theta, \phi \in \mathbb{T}$ with geodesic distance $|\theta - \phi| \leq \pi$,

$$|P(\theta) - P(\phi)| \leq r \|P\|_\infty |\theta - \phi|.$$

3 Finite-section symbol estimates

The next lemma is the common analytic input behind both Theorem 1.5 and Theorem 1.6. The constants are not optimized, but the estimate is explicit.

Lemma 3.1 (Window estimate with modulus of continuity). *Let p be a real-valued trigonometric polynomial, and let $h \in \mathbb{N}$ satisfy $2h - 1 \leq d$. Then*

$$\|T_d(p)\|_{\text{op}} \geq \|p\|_\infty - \frac{3\pi}{2h} \|p'\|_\infty.$$

More generally, if $\omega_p(t) = \sup_{|\theta - \phi| \leq t} |p(\theta) - p(\phi)|$, then

$$\|T_d(p)\|_{\text{op}} \geq \|p\|_\infty - \int_{\mathbb{T}} \omega_p(|\phi|) W_h(\phi) \frac{d\phi}{2\pi},$$

where W_h is the probability kernel defined in the proof below and satisfies $\int_{\mathbb{T}} |\phi| W_h(\phi) d\phi / (2\pi) \leq 3\pi/(2h)$.

Proof. Choose an interval of $2h - 1$ consecutive indices inside $\{1, \dots, d\}$, centered at an integer x_0 , and define the triangular window

$$u_x = \max(0, h - |x - x_0|), \quad x = 1, \dots, d.$$

For $\theta \in \mathbb{T}$, set $v_\theta(x) = u_x e^{ix\theta}$. A direct calculation gives

$$\frac{\langle T_d(p) v_\theta, v_\theta \rangle}{\|u\|_2^2} = (p * W_h)(\theta),$$

where $W_h \geq 0$ is the probability kernel whose Fourier coefficients are the normalized autocorrelations

$$\widehat{W}_h(k) = \frac{\sum_x u_x u_{x+k}}{\sum_x u_x^2}.$$

Therefore

$$\|T_d(p)\|_{\text{op}} \geq \sup_{\theta \in \mathbb{T}} |(p * W_h)(\theta)|. \quad (1)$$

We need a first-moment bound for W_h . Let

$$F_h(\phi) = \frac{1}{h} \left| \sum_{r=0}^{h-1} e^{ir\phi} \right|^2 = \frac{1}{h} \frac{\sin^2(h\phi/2)}{\sin^2(\phi/2)}$$

be the Fejer kernel. Up to a harmless phase, the Fourier transform of the triangular window is hF_h , hence

$$W_h(\phi) = \frac{h^2 F_h(\phi)^2}{\|u\|_2^2}.$$

Moreover

$$\|u\|_2^2 = h^2 + 2 \sum_{j=1}^{h-1} j^2 = \frac{2h^3 + h}{3}.$$

Identifying $\mathbb{T}$ with $[-\pi, \pi]$ and using $|\sin(\phi/2)| \geq |\phi|/\pi$ on this interval,

$$F_h(\phi) \leq \min \left(h, \frac{\pi^2}{h\phi^2} \right), \quad 0 < |\phi| \leq \pi.$$

Consequently,

$$\begin{aligned} \int_{\mathbb{T}} |\phi| F_h(\phi)^2 \frac{d\phi}{2\pi} &= \frac{1}{\pi} \int_0^\pi \phi F_h(\phi)^2 d\phi \\ &\leq \frac{1}{\pi} \left(\int_0^{\pi/h} \phi h^2 d\phi + \int_{\pi/h}^\pi \phi \frac{\pi^4}{h^2 \phi^4} d\phi \right) \\ &\leq \pi. \end{aligned}$$

It follows that

$$M_1(W_h) := \int_{\mathbb{T}} |\phi| W_h(\phi) \frac{d\phi}{2\pi} \leq \frac{h^2}{\|u\|_2^2} \pi = \frac{3h\pi}{2h^2 + 1} \leq \frac{3\pi}{2h}. \quad (2)$$

Choose θ_* with $|p(\theta_*)| = \|p\|_\infty$, and multiply p by -1 if necessary so that $p(\theta_*) = \|p\|_\infty$. Then

$$p(\theta_* - \phi) \geq p(\theta_*) - \omega_p(|\phi|).$$

Since W_h is a probability kernel,

$$(p * W_h)(\theta_*) \geq \|p\|_\infty - \int_{\mathbb{T}} \omega_p(|\phi|) W_h(\phi) \frac{d\phi}{2\pi}.$$

Together with (1), this proves the modulus version. The Lipschitz version follows from $\omega_p(t) \leq \|p'\|_\infty t$ and (2). $\square$

Lemma 3.2 (Quantitative symbol transfer). *Let $w \geq 1$, and let p be a real-valued trigonometric polynomial of degree at most w . If $h \in \mathbb{N}$, $h > w$, and $2h - 1 \leq d$, then*

$$\|T_d(p)\|_{\text{op}} \geq \left(1 - \frac{3\pi w}{2h} \right) \|p\|_\infty.$$

In particular, if $d \geq 20w$, then

$$\frac{1}{2} \|p\|_\infty \leq \|T_d(p)\|_{\text{op}} \leq \|p\|_\infty.$$

Proof. The lower bound follows from Lemma 3.1 and Bernstein's inequality $\|p'\|_\infty \leq w \|p\|_\infty$. For the headline bound, choose $h = 10w$. Then $h > w$, $2h - 1 \leq 20w - 1 \leq d$, and $1 - 3\pi/20 > 1/2$. The upper bound is Lemma 2.2. $\square$

4 Banded and tame Toeplitz Spencer

We first prove Theorem 1.5.

Proof of Theorem 1.5. Write $A_i = T_d(p_i)$, where p_i is real-valued and has degree at most w . By Lemma 3.2,

$$\|p_i\|_\infty \leq 2 \|T_d(p_i)\|_{\text{op}} \leq 2.$$

Let $N_g = 8w$ and let $\theta_j = 2\pi j/N_g$, $j = 0, \dots, N_g - 1$. If P is any trigonometric polynomial of degree at most w , then every point of $\mathbb{T}$ is within distance π/N_g of some θ_j . By Lemma 2.3,

$$\|P\|_\infty \leq \max_{j < N_g} |P(\theta_j)| + \frac{\pi}{N_g} w \|P\|_\infty = \max_{j < N_g} |P(\theta_j)| + \frac{\pi}{8} \|P\|_\infty.$$

Thus

$$\|P\|_\infty \leq \left(1 - \frac{\pi}{8}\right)^{-1} \max_{j < N_g} |P(\theta_j)| \leq 1.66 \max_{j < N_g} |P(\theta_j)|. \quad (3)$$

For each i , form the vector

$$x_i = (p_i(\theta_0), \dots, p_i(\theta_{N_g-1})) \in \mathbb{C}^{N_g}.$$

Then $\|x_i\|_\infty \leq 2$. Applying Theorem 2.1 to $x_i/2$ gives signs $\varepsilon \in \{-1, 1\}^n$ such that

$$\max_{j < N_g} \left| \sum_{i=1}^n \varepsilon_i p_i(\theta_j) \right| \leq 2C_S \sqrt{n \log \left(e + \frac{8w}{n} \right)}.$$

Since $\log(e + 8x) \leq C_0 \log(e + x)$ for an absolute constant C_0 and all $x \geq 0$, this is

$$\max_{j < N_g} \left| \sum_i \varepsilon_i p_i(\theta_j) \right| \leq C' \sqrt{n \log \left(e + \frac{w}{n} \right)}.$$

Set $P = \sum_i \varepsilon_i p_i$. By (3) and Lemma 2.2,

$$\left\| \sum_{i=1}^n \varepsilon_i A_i \right\|_{\text{op}} = \|T_d(P)\|_{\text{op}} \leq \|P\|_\infty \leq 1.66 C' \sqrt{n \log \left(e + \frac{w}{n} \right)}.$$

If $w \leq n$, then $\log(e + w/n)$ is bounded by an absolute constant. $\square$

Proof of Theorem 1.6. Assume first that $d \geq 2$. Choose $h = \lfloor (d+1)/2 \rfloor$. Then $2h - 1 \leq d$ and $h \geq d/2$. Lemma 3.1 gives, for each i ,

$$\|p_i\|_\infty \leq \|T_d(p_i)\|_{\text{op}} + \frac{3\pi}{2h} \|p'_i\|_\infty \leq 1 + \frac{3\pi L}{d}.$$

Let $B = 1 + 3\pi L/d$. Use the grid $\theta_j = 2\pi j/(8d)$, $j = 0, \dots, 8d - 1$. Since every signed sum $P = \sum_i \varepsilon_i p_i$ has degree at most $d - 1$, the same Bernstein-grid argument as above, with w replaced by d , gives

$$\|P\|_\infty \leq 1.66 \max_{0 \leq j < 8d} |P(\theta_j)|.$$

The vectors $x_i = (p_i(\theta_j))_{j < 8d}$ satisfy $\|x_i\|_\infty \leq B$. Applying Theorem 2.1 to x_i/B gives signs such that

$$\max_{j < 8d} \left| \sum_i \varepsilon_i p_i(\theta_j) \right| \leq BC_S \sqrt{n \log \left(e + \frac{8d}{n} \right)}.$$

The grid inequality and Lemma 2.2 imply

$$\left\| \sum_i \varepsilon_i A_i \right\|_{\text{op}} \leq CB \sqrt{n \log \left(e + \frac{d}{n} \right)} \leq C \left(1 + \frac{L}{d} \right) \sqrt{n \log \left(e + \frac{d}{n} \right)}.$$

The case $d = 1$ is exactly the scalar Spencer problem and is absorbed by changing the universal constant. If $d \leq n$ and $L = O(d)$, the logarithm is bounded by an absolute constant. $\square$

Proof of Proposition 1.7. Repeat the preceding proof without the finite-section estimate. The vectors $x_i = (p_i(2\pi j/(8d)))_{j < 8d}$ have ℓ_∞ norm at most K , and the grid inequality for degree $< d$ trigonometric polynomials gives the claimed bound. $\square$

5 Scale-localized recovery and the top-band reduction

This section adds a second, independent reduction for full-bandwidth Toeplitz sections. The banded proof used that the whole symbol is visible to the finite section. Here we show that this visibility holds after frequency localization at all scales below the top part of the spectrum.

For $K \in L^1(\mathbb{T})$ write

$$(p * K)(\theta) = \int_{\mathbb{T}} p(\theta - \phi) K(\phi) \frac{d\phi}{2\pi}, \quad \widehat{K}(k) = \int_{\mathbb{T}} K(\phi) e^{-ik\phi} \frac{d\phi}{2\pi}.$$

Lemma 5.1 (Toeplitz Schur transference). *For every $K \in L^1(\mathbb{T})$ and every trigonometric polynomial p of degree $< d$,*

$$\|T_d(p * K)\|_{\text{op}} \leq \|K\|_{L^1(\mathbb{T})} \|T_d(p)\|_{\text{op}}.$$

This is the elementary L^1 half of Bennett's Toeplitz Schur multiplier theorem [6]; we include the proof.

Proof. Write $p(\theta) = \sum_{|k| < d} c_k e^{ik\theta}$. Then $p * K$ has coefficient $c_k \widehat{K}(k)$. Let

$$D_\phi = \text{diag}(e^{ix\phi})_{x=1}^d.$$

Since

$$(D_\phi^* T_d(p) D_\phi)_{xy} = c_{x-y} e^{-i(x-y)\phi},$$

we have the entrywise Bochner identity

$$T_d(p * K) = \int_{\mathbb{T}} D_\phi^* T_d(p) D_\phi K(\phi) \frac{d\phi}{2\pi}.$$

The triangle inequality and unitary invariance give the claimed bound. $\square$

Let F_M denote the Fejér kernel,

$$\hat{F}_M(k) = \max\left(0, 1 - \frac{|k|}{M}\right), \quad \|F_M\|_{L^1} = 1.$$

For $m \geq 1$ set

$$V_m = 2F_{2m} - F_m.$$

Then $\hat{V}_m(k) = 1$ for $|k| \leq m$, $\hat{V}_m(k) = 2 - |k|/m$ for $m \leq |k| \leq 2m$, and $\hat{V}_m(k) = 0$ for $|k| \geq 2m$; moreover $\|V_m\|_{L^1} \leq 3$. Define the low-pass and dyadic-band operators

$$L_m p = p * V_m, \quad \Pi_j p = p * (V_{2^j} - V_{2^{j-1}}) \quad (j \geq 1).$$

Thus $L_m p$ has degree at most $2m$, while $\Pi_j p$ is supported in $2^{j-1} < |k| < 2^{j+1}$ and is generated by a kernel of L^1 norm at most 6.

Lemma 5.2 (Scale-localized symbol recovery). *Let p be real-valued of degree $< d$.*

(i) *If $504m \leq d$, then*

$$\|L_m p\|_\infty \leq 6 \|T_d(p)\|_{\text{op}}.$$

(ii) *If $504 \cdot 2^j \leq d$, then*

$$\|\Pi_j p\|_\infty \leq 12 \|T_d(p)\|_{\text{op}}.$$

Proof. For (i), the degree of $L_m p$ is at most $2m$ and $d \geq 504m \geq 20(2m)$, so Lemma 3.2 gives

$$\|L_m p\|_\infty \leq 2 \|T_d(L_m p)\|_{\text{op}}.$$

By Lemma 5.1,

$$\|T_d(L_m p)\|_{\text{op}} \leq \|V_m\|_{L^1} \|T_d(p)\|_{\text{op}} \leq 3 \|T_d(p)\|_{\text{op}}.$$

Combining the two inequalities proves (i). The proof of (ii) is identical: $\Pi_j p$ has degree at most 2^{j+1} , the condition gives $d \geq 20 \cdot 2^{j+1}$, and the defining kernel has L^1 norm at most 6. $\square$

Define the finite-section symbol-recovery constants

$$\kappa(d) = \sup\{\|p\|_\infty : p \text{ real-valued, } \deg p < d, \|T_d(p)\|_{\text{op}} \leq 1\}.$$

For $d \geq 504$ let $m_0 = \lfloor d/504 \rfloor$ and define

$$\kappa_{\text{top}}(d) = \sup\{\|h\|_\infty : h \text{ real-valued, } \hat{h}(k) = 0 \text{ for } |k| \leq m_0, \deg h < d, \|T_d(h)\|_{\text{op}} \leq 1\},$$

and let $\kappa_{\text{band}}(d)$ be the same supremum with the spectrum of h further confined to one dyadic band $2^{j-1} \leq |k| \leq 2^{j+1}$ satisfying $2^{j+1} > m_0$.

Proposition 5.3 (The recovery obstruction is top-band localized). *For all $d \geq 504$,*

$$\kappa(d) \leq 6 + 4\kappa_{\text{top}}(d), \quad \kappa_{\text{top}}(d) \leq 78\kappa_{\text{band}}(d).$$

Consequently,

$$\kappa(d) \leq 6 + 312\kappa_{\text{band}}(d).$$

In particular, if $\kappa(d)$ is unbounded, then the obstruction is already witnessed on symbols supported in a single dyadic band contained in $[d/4032, 2d]$.

Proof. Let p be admissible for $\kappa(d)$, so $\deg p < d$ and $\|T_d(p)\|_{\text{op}} \leq 1$. Split

$$p = \ell + h, \quad \ell = L_{m_0}p, \quad h = p - L_{m_0}p.$$

By Lemma 5.2, $\|\ell\|_{\infty} \leq 6$. Moreover, Lemma 5.1 gives

$$\|T_d(\ell)\|_{\text{op}} \leq \|V_{m_0}\|_{L^1} \|T_d(p)\|_{\text{op}} \leq 3,$$

so

$$\|T_d(h)\|_{\text{op}} \leq \|T_d(p)\|_{\text{op}} + \|T_d(\ell)\|_{\text{op}} \leq 4.$$

Since h has no frequencies $|k| \leq m_0$, the definition of $\kappa_{\text{top}}(d)$ gives

$$\|p\|_{\infty} \leq \|\ell\|_{\infty} + \|h\|_{\infty} \leq 6 + 4\kappa_{\text{top}}(d).$$

Taking the supremum over admissible p proves the first inequality.

For the second inequality, let h be admissible for $\kappa_{\text{top}}(d)$. Let $J = \lceil \log_2 d \rceil$. The de la Vallée Poussin decomposition telescopes to

$$h = h * V_1 + \sum_{j=1}^J \Pi_j h,$$

because V_{2^j} is identically one on the spectrum of h . The term $h * V_1$ can occur only when $m_0 = 1$; all nonzero terms have spectra in dyadic bands with top edge $> m_0$ and lower edge $< d$. Since $d/m_0 \leq 1008$, there are at most 13 such pieces. For each piece q , the kernel producing it has L^1 norm at most 6, hence Lemma 5.1 gives $\|T_d(q)\|_{\text{op}} \leq 6$. Therefore

$$\|h\|_{\infty} \leq \sum_q \|q\|_{\infty} \leq 13 \cdot 6 \kappa_{\text{band}}(d),$$

which proves the second inequality. The relevant bands lie in $[d/4032, 2d]$ because $m_0 \geq d/1008$ and every such band has lower edge at least one quarter of its top edge. $\square$

Corollary 5.4 (Discrepancy under bounded top-band recovery). *For arbitrary self-adjoint Toeplitz matrices $A_i = T_d(p_i)$ with $\|A_i\|_{\text{op}} \leq 1$,*

$$\text{disc}(A_1, \dots, A_n) \leq C\kappa(d) \sqrt{n \log \left(e + \frac{d}{n} \right)} \leq C(1 + \kappa_{\text{band}}(d)) \sqrt{n \log \left(e + \frac{d}{n} \right)}.$$

Consequently, if $\kappa_{\text{band}}(d) = O(1)$ uniformly in d , then full-bandwidth Toeplitz matrices satisfy the Matrix Spencer bound $C\sqrt{n}$ whenever $d \leq n$.

Proof. By the definition of $\kappa(d)$, each symbol $p_i/\kappa(d)$ has L^∞ norm at most 1. Proposition 1.7 applied to these normalized symbols gives the first bound. Proposition 5.3 gives the second after absorbing constants. $\square$

Remark 5.5 (Why frequency sparsity alone is not used here). Under the coefficient convention $T_d(p) = (c_{x-y})_{x,y=1}^d$, a single Fourier frequency is a single diagonal shift, not a rank-one or rank-two matrix. For example, the self-adjoint frequency $2 \cos(k\theta)$ gives the two k -th off-diagonals and typically has rank comparable to d when $k \ll d$. Thus sparse frequency support is not, by itself, a low-rank hypothesis for finite Toeplitz sections. The reduction above uses only frequency localization and Schur transference, avoiding any rank claim for individual frequencies.

Remark 5.6 (Constructive signs). The proofs of Theorems 1.5 and 1.6 are algorithmic modulo the chosen implementation of vector Spencer. Given the Toeplitz coefficients, one samples the relevant grid and applies any constructive Spencer algorithm to the resulting scalar matrix. The same signs then satisfy the matrix bound by the argument above.

6 Finite-entropy lifts and full Toeplitz/Hankel signing

This section proves the abstract finite-lift principle and then applies it to arbitrary scalar Toeplitz and Hankel matrices. The key point is that a finite Toeplitz contraction has a bounded L^∞ inducer, and after de la Vallée Poussin smoothing the relevant Fourier data live in only $O(d)$ scalar samples.

6.1 The block-lift theorem

We use the following block-diagonal form of the matrix discrepancy theorem of Dadush, Jiang and Reis [13]. The nonuniform version below follows by grouping blocks by size and absorbing absolute constants.

Theorem 6.1 (Block-diagonal Matrix Spencer). *There is a universal constant C_{BD} such that the following holds. Let $Y_1, \dots, Y_n$ be self-adjoint elements of*

$$\mathcal{B} = \bigoplus_{\alpha=1}^m M_{r_\alpha}(\mathbb{C}), \quad \|Y\|_{\mathcal{B}} := \max_{\alpha} \|Y_{\alpha}\|_{\text{op}},$$

with $\|Y_i\|_{\mathcal{B}} \leq 1$. Put $r_ := \max_{\alpha} r_{\alpha}$ and $R := \sum_{\alpha} r_{\alpha}$. Then there are signs $\varepsilon \in \{-1, 1\}^n$ such that*

$$\left\| \sum_{i=1}^n \varepsilon_i Y_i \right\|_{\mathcal{B}} \leq C_{\text{BD}} \sqrt{n \log \left(e + \frac{r_* R}{n} \right)}.$$

The same estimate holds for arbitrary complex block-diagonal matrices after replacing each block by its Hermitian dilation.

Definition 6.2 (Finite block lift). Let $A_1, \dots, A_n \in M_D(\mathbb{C})$ and let $\mathcal{B} = \bigoplus_{\alpha=1}^m M_{r_\alpha}(\mathbb{C})$. We say that the tuple admits a K -bounded lift to $\mathcal{B}$ if there are elements $Y_i \in \mathcal{B}$ and a linear map Π defined on $\text{span}\{Y_1, \dots, Y_n\}$ such that

$$\Pi(Y_i) = A_i, \quad \|\Pi(Y)\|_{\text{op}} \leq \|Y\|_{\mathcal{B}} \quad (Y \in \text{span}\{Y_1, \dots, Y_n\}),$$

and $\|Y_i\|_{\mathcal{B}} \leq K \|A_i\|_{\text{op}}$ for every i .

Theorem 6.3 (Finite block-lift Matrix Spencer). *There is a universal constant C with the following property. Let $A_1, \dots, A_n \in M_D(\mathbb{C})$ satisfy $\|A_i\|_{\text{op}} \leq 1$, and suppose the tuple admits a K -bounded lift to $\mathcal{B} = \bigoplus_{\alpha=1}^m M_{r_\alpha}(\mathbb{C})$. With $r_* := \max_{\alpha} r_{\alpha}$, $R := \sum_{\alpha} r_{\alpha}$, and $\eta(\mathcal{B}) := r_* R$,*

$$\text{disc}_{\text{op}}(A_1, \dots, A_n) \leq CK \sqrt{n \log \left(e + \frac{\eta(\mathcal{B})}{n} \right)}.$$

If the A_i are self-adjoint and the lifts may be chosen self-adjoint, the same estimate holds for disc.

Proof. First assume the lifts are self-adjoint. Applying Theorem 6.1 to Y_i/K gives signs with

$$\left\| \sum_i \varepsilon_i Y_i \right\|_{\mathcal{B}} \leq CK \sqrt{n \log \left(e + \frac{r_* R}{n} \right)}.$$

Since Π is linear and contractive on the span of the lifts,

$$\left\| \sum_i \varepsilon_i A_i \right\|_{\text{op}} = \left\| \Pi \left(\sum_i \varepsilon_i Y_i \right) \right\|_{\text{op}} \leq \left\| \sum_i \varepsilon_i Y_i \right\|_{\mathcal{B}}.$$

For arbitrary complex lifts, replace each block $Y_{i,\alpha}$ by its Hermitian dilation $\begin{pmatrix} 0 & Y_{i,\alpha} \\ Y_{i,\alpha}^* & 0 \end{pmatrix}$. This doubles all block sizes and changes only the universal constant, while preserving the norm of every signed sum. $\square$

Definition 6.4 (Sampled multiplier lift). Let Ω be compact and let $\mathcal{F} \subset C(\Omega; M_b(\mathbb{C}))$ be a finite-dimensional linear space. A finite set $Q \subset \Omega$ is L -norming for $\mathcal{F}$ if

$$\|F\|_{L^\infty(\Omega; M_b)} \leq L \max_{\omega \in Q} \|F(\omega)\|_{\text{op}} \quad (F \in \mathcal{F}).$$

A tuple $A_1, \dots, A_n$ admits a (K, L, Q, b) sampled multiplier lift if there are $F_i \in \mathcal{F}$ and a fixed linear contraction $\Pi : \text{span}\{F_1, \dots, F_n\} \rightarrow M_D(\mathbb{C})$ such that $\Pi(F_i) = A_i$ and $\|F_i\|_{L^\infty} \leq K \|A_i\|_{\text{op}}$.

Theorem 6.5 (Sampled multiplier Matrix Spencer). *If $A_1, \dots, A_n$ admit a (K, L, Q, b) sampled multiplier lift and $\|A_i\|_{\text{op}} \leq 1$, then*

$$\text{disc}_{\text{op}}(A_1, \dots, A_n) \leq CKL \sqrt{n \log \left(e + \frac{b^2 |Q|}{n} \right)}.$$

For $b = 1$ this is the vector-Spencer bound with $|Q|$ scalar samples.

Proof. Evaluate each F_i on Q and form

$$Y_i = \bigoplus_{\omega \in Q} F_i(\omega) \in \bigoplus_{\omega \in Q} M_b(\mathbb{C}).$$

The block algebra has $r_* = b$ and $R = b|Q|$, while $\|Y_i\| \leq K$. Theorem 6.3 gives signs with

$$\max_{\omega \in Q} \left\| \sum_i \varepsilon_i F_i(\omega) \right\|_{\text{op}} \leq CK \sqrt{n \log \left(e + \frac{b^2 |Q|}{n} \right)}.$$

The L -norming property and the contraction Π give the result. $\square$

Lemma 6.6 (Norming grid for matrix trigonometric polynomials). *Fix $r \geq 1$. Let $P : \mathbb{T}^r \rightarrow M_b(\mathbb{C})$ be a matrix-valued trigonometric polynomial with coordinate degrees at most $N_1, \dots, N_r$. For each j , let G_j be a uniform grid in $\mathbb{T}$ of size at least $8rN_j$ when $N_j \geq 1$ and of size 1 when $N_j = 0$. Put $G = G_1 \times \dots \times G_r$. Then*

$$\|P\|_{L^\infty(\mathbb{T}^r; M_b)} \leq 2 \max_{\theta \in G} \|P(\theta)\|_{\text{op}}.$$

In particular, $|G| = O_r(\prod_{j=1}^r (1 + N_j))$.

Proof. The Banach-valued Bernstein inequality gives $\|\partial_j P\|_\infty \leq N_j \|P\|_\infty$ for each coordinate. Indeed, apply the scalar Bernstein inequality to $\varphi(P(\cdot))$ for norm-one linear functionals φ on $M_b(\mathbb{C})$ and then take the supremum over such φ . Choose θ^* where $\|P(\theta^*)\|$ is within δ of $\|P\|_\infty$, and choose $\theta \in G$ with $|\theta_j - \theta_j^*| \leq \pi/|G_j|$. Then

$$\|P(\theta^*) - P(\theta)\| \leq \sum_{j=1}^r \frac{\pi N_j}{|G_j|} \|P\|_\infty \leq \frac{\pi}{8} \|P\|_\infty.$$

Letting $\delta \downarrow 0$ gives the claim, after increasing the numerical constant to 2. $\square$

6.2 Full scalar Toeplitz and Hankel

For a finite sequence $(a_k)_{-(d-1) \leq k \leq d-1}$, write

$$T_d(a) = (a_{r-s})_{0 \leq r, s < d}.$$

A function $\phi \in L^\infty(\mathbb{T})$ induces $T_d(a)$ if $\widehat{\phi}(k) = a_k$ for $|k| < d$.

Theorem 6.7 (Finite Toeplitz extension theorem). *There is an absolute constant K_T such that every complex $d \times d$ Toeplitz matrix A has an inducer $\phi_A \in L^\infty(\mathbb{T})$ satisfying*

$$\widehat{\phi_A}(k) = a_k \quad (|k| < d), \quad \|\phi_A\|_{L^\infty} \leq K_T \|A\|_{\text{op}}.$$

In the self-adjoint case the inducer may be chosen real-valued. In the scalar self-adjoint case one may take an absolute constant independent of d ; this is the finite Toeplitz extension theorem of Nikolskaya–Farforovskaya, in the form recorded and developed by Courtney–Sarason [24, 12].

Let $\mathcal{V}_N$ be a de la Vallée Poussin kernel with

$$\widehat{\mathcal{V}_N}(k) = 1 \quad (|k| \leq N), \quad \widehat{\mathcal{V}_N}(k) = 0 \quad (|k| > 2N), \quad \|\mathcal{V}_N\|_{L^1} \leq C_V.$$

Theorem 6.8 (Arbitrary scalar Toeplitz signing). *There is a universal constant C such that, for every collection of arbitrary complex Toeplitz matrices $A_1, \dots, A_n \in M_d(\mathbb{C})$ with $\|A_i\|_{\text{op}} \leq 1$,*

$$\text{disc}_{\text{op}}(A_1, \dots, A_n) \leq C \sqrt{n \log \left(e + \frac{d}{n} \right)}.$$

In particular, self-adjoint Toeplitz matrices satisfy Matrix Spencer in the regime $d \leq n$.

Proof. For each i , choose an inducer ϕ_i from Theorem 6.7. Put $p_i = \phi_i * \mathcal{V}_{d-1}$. Then p_i is a trigonometric polynomial of degree at most $2d-2$, $\widehat{p_i}(k) = \widehat{\phi_i}(k)$ for $|k| < d$, and $\|p_i\|_\infty \leq C_V K_T$. Hence $T_d(p_i) = A_i$.

Choose a uniform grid $Q \subset \mathbb{T}$ of size $O(d)$ that is 2-norming for trigonometric polynomials of degree at most $2d$; this is Lemma 6.6 with $r = b = 1$. Applying Theorem 6.5 with $b = 1$, $K = C_V K_T$, and $L = 2$, gives signs such that

$$\left\| \sum_i \varepsilon_i p_i \right\|_\infty \leq C \sqrt{n \log \left(e + \frac{d}{n} \right)}.$$

Since Toeplitz compression is contractive from $L^\infty(\mathbb{T})$ to operator norm by Lemma 2.2,

$$\left\| \sum_i \varepsilon_i A_i \right\|_{\text{op}} = \left\| T_d \left(\sum_i \varepsilon_i p_i \right) \right\|_{\text{op}} \leq \left\| \sum_i \varepsilon_i p_i \right\|_{\infty}.$$

□

Corollary 6.9 (Arbitrary scalar Hankel signing). *For arbitrary complex Hankel matrices $H_1, \dots, H_n \in M_d(\mathbb{C})$ with $\|H_i\|_{\text{op}} \leq 1$,*

$$\text{disc}_{\text{op}}(H_1, \dots, H_n) \leq C \sqrt{n \log \left(e + \frac{d}{n} \right)}.$$

Thus self-adjoint Hankel matrices satisfy Matrix Spencer whenever $d \leq n$.

Proof. Let J_d be the reversal matrix. If H is Hankel, then HJ_d is Toeplitz and $\|HJ_d\|_{\text{op}} = \|H\|_{\text{op}}$. Apply Theorem 6.8 to the Toeplitz matrices $H_i J_d$ and use

$$\left\| \sum_i \varepsilon_i H_i \right\|_{\text{op}} = \left\| \left(\sum_i \varepsilon_i H_i J_d \right) J_d \right\|_{\text{op}} = \left\| \sum_i \varepsilon_i H_i J_d \right\|_{\text{op}}.$$

□

Remark 6.10 (Relation to the banded and top-band refinements). Theorem 6.8 proves the optimal Toeplitz Spencer scale for scalar Toeplitz contractions by using a bounded extension. The banded and top-band results are still useful because they work directly with the finite-section polynomial symbol. In particular, Theorem 1.5 recovers the smaller entropy w for bandwidth- w symbols, while Section 5 localizes any finite-section-only symbol-recovery loss to the top dyadic bands.

6.3 Block Toeplitz, finite-dimensional algebras, and group convolution

A $b \times b$ matrix-valued Toeplitz matrix is a Toeplitz matrix whose entries are $b \times b$ blocks. More generally, an r -level block Toeplitz matrix is indexed by a box $\prod_{j=1}^r \{0, \dots, d_j - 1\}$ and has entries depending only on coordinate differences. Its total dimension is $D = b \prod_{j=1}^r d_j$.

Theorem 6.11 (Fixed-level block Toeplitz and Hankel). *Fix $r \geq 1$ and $b \geq 1$. There is a constant $C_{r,b}$ such that every collection of r -level $b \times b$ block Toeplitz contractions $A_1, \dots, A_n \in M_D(\mathbb{C})$ satisfies*

$$\text{disc}_{\text{op}}(A_1, \dots, A_n) \leq C_{r,b} \sqrt{n \log \left(e + \frac{D}{n} \right)}.$$

The same bound holds for the corresponding r -level block Hankel class. Hence, for fixed r, b and $D \leq n$, these classes satisfy the Matrix Spencer bound $O_{r,b}(\sqrt{n})$.

Proof. The scalar multilevel Toeplitz extension theorem, in the fixed-level form following from the work of Carlsson and Perfekt [11], gives a bounded extension of a finite multilevel Toeplitz matrix to an $L^\infty(\mathbb{T}^r)$ multiplier, with constants depending only on r . Applying this entrywise to the fixed $b \times b$ block entries gives an $M_b(\mathbb{C})$ -valued inducer with loss depending only on r and b .

Convolution with a product de la Vallée Poussin kernel gives a matrix trigonometric polynomial with coordinate degrees $O(d_j)$ and with the same finite Toeplitz coefficients. Lemma 6.6 gives a norming grid of size $O_r(\prod_j d_j) = O_r(D/b)$. Theorem 6.5 with block size b gives the claimed bound after absorbing r, b into the constant. The Hankel case follows by multiplying by the tensor-product reversal matrix. $\square$

Theorem 6.12 (Common finite-dimensional C^* -algebra). *Let $\mathcal{A} \subset M_D(\mathbb{C})$ be a finite-dimensional C^* -algebra represented as*

$$\mathcal{A} \simeq \bigoplus_{\alpha=1}^m M_{r_\alpha}(\mathbb{C}) \otimes I_{s_\alpha}.$$

Let $r_ := \max_\alpha r_\alpha$ and $R := \sum_\alpha r_\alpha$. If $A_1, \dots, A_n \in \mathcal{A}$ satisfy $\|A_i\|_{\text{op}} \leq 1$, then*

$$\text{disc}_{\text{op}}(A_1, \dots, A_n) \leq C \sqrt{n \log \left(e + \frac{r_* R}{n} \right)}.$$

If the matrices are self-adjoint, this is a self-adjoint discrepancy bound.

Proof. An element of $\mathcal{A}$ is unitarily equivalent to $\bigoplus_\alpha (B_\alpha \otimes I_{s_\alpha})$. Delete the multiplicity identities to obtain an isometric lift in $\bigoplus_\alpha M_{r_\alpha}(\mathbb{C})$, and represent back by $\bigoplus_\alpha B_\alpha \mapsto \bigoplus_\alpha (B_\alpha \otimes I_{s_\alpha})$. This is an isometric linear map on the reduced algebra. Apply Theorem 6.3 with $K = 1$. $\square$

Corollary 6.13 (Diagonal, bounded-block, and association-scheme examples). *Simultaneously diagonalizable normal contractions satisfy*

$$\text{disc}_{\text{op}}(A_1, \dots, A_n) \leq C \sqrt{n \log \left(e + \frac{D}{n} \right)}.$$

If all A_i are simultaneously unitarily block diagonal with common block sizes at most b , then

$$\text{disc}_{\text{op}}(A_1, \dots, A_n) \leq C \sqrt{n \log \left(e + \frac{bD}{n} \right)}.$$

In particular, fixed block size gives $O_b(\sqrt{n})$ for $D \leq n$. If the A_i lie in a commutative Bose–Mesner algebra with s primitive idempotents, then $\text{disc}_{\text{op}}(A_1, \dots, A_n) \leq C \sqrt{n \log(e + s/n)}$.

Theorem 6.14 (Finite group convolution). *Let G be finite, let $\widehat{G}$ be its irreducible unitary dual, put $q(G) = \max_{\pi \in \widehat{G}} \dim \pi$ and $S(G) = \sum_{\pi \in \widehat{G}} \dim \pi$. If $L_{f_1}, \dots, L_{f_n}$ are left convolution operators on $\ell^2(G)$ with $\|L_{f_i}\|_{\text{op}} \leq 1$, then*

$$\text{disc}_{\text{op}}(L_{f_1}, \dots, L_{f_n}) \leq C \sqrt{n \log \left(e + \frac{q(G)S(G)}{n} \right)}.$$

Consequently, for finite abelian groups,

$$\text{disc}_{\text{op}} \leq C \sqrt{n \log \left(e + \frac{|G|}{n} \right)},$$

and for any family of groups with $q(G) \leq q$ fixed, arbitrary convolution kernels satisfy $O_q(\sqrt{n})$ whenever $|G| \leq n$.

Proof. The finite Fourier transform identifies the left convolution algebra with the reduced block algebra $\bigoplus_{\pi \in \widehat{G}} M_{d_\pi}(\mathbb{C})$, where $d_\pi = \dim \pi$, with operator norm $\max_\pi \|\widehat{f}(\pi)\|_{\text{op}}$. Theorem 6.3 gives the displayed bound with $r_* = q(G)$ and $R = S(G)$. If G is abelian then every $d_\pi = 1$ and $S(G) = |G|$. If $q(G) \leq q$, then $q(G)S(G) \leq q|G|$ because $S(G) \leq \sum_\pi d_\pi^2 = |G|$. $\square$

Remark 6.15 (Relation to recent algebraic and group-representation results). The common- C^* -algebra statement is closely related to the algebraic Matrix Spencer theorem of Akbas and Sra [1]. The finite-group convolution theorem above signs arbitrary convolution kernels and should be distinguished from the theorem of Bandeira and Bölcskei, which signs the specific list of left-regular representation matrices indexed by group elements with a universal $C\sqrt{|G|}$ bound [5].

6.4 Closure operations, random inputs, and the boundary of finite lifts

Proposition 6.16 (Compressions, quotients, bounded sums, and tensor amplification). *The finite-lift class is stable under the following operations.*

- (i) *If $A_i = V^* B_i V$ for a fixed isometry V , then any discrepancy bound for the B_i holds for the A_i .*
- (ii) *If $A_i = \Phi(B_i)$ for a fixed linear map with $\|\Phi\| \leq L$, then the bound is multiplied by L .*
- (iii) *If $A_i = A_i^{(1)} + \dots + A_i^{(s)}$ and the ℓ -th component has lift loss K_ℓ to a block algebra $\mathcal{B}_\ell$, then the tuple has lift loss at most $\sum_\ell K_\ell$ to $\bigoplus_\ell \mathcal{B}_\ell$.*
- (iv) *Tensoring a solved class with a fixed $c \times c$ matrix factor multiplies the entropy parameter by c^2 and preserves the Spencer-regime conclusion for fixed c .*

Consequently, if $A_i = T_i + H_i$ with T_i Toeplitz, H_i Hankel, and $\|T_i\|_{\text{op}} + \|H_i\|_{\text{op}} \leq K$, then

$$\text{disc}_{\text{op}}(A_1, \dots, A_n) \leq CK \sqrt{n \log \left(e + \frac{d}{n} \right)}.$$

Proof. The first two assertions follow from applying the same signs and using $\|V^*(\sum_i \varepsilon_i B_i)V\| \leq \|\sum_i \varepsilon_i B_i\|$ and $\|\Phi(\sum_i \varepsilon_i B_i)\| \leq L \|\sum_i \varepsilon_i B_i\|$. For sums, take the direct sum of the component lifts and define the reconstruction map by adding the reconstructed components; its norm on the relevant span is at most $\sum_\ell K_\ell$. For tensor amplification, replace each block $M_r(\mathbb{C})$ by $M_{cr}(\mathbb{C})$; both r_* and R are multiplied by c , hence η is multiplied by c^2 . The Toeplitz-plus-Hankel statement combines Theorem 6.8, Corollary 6.9, and the bounded-sum assertion. $\square$

Corollary 6.17 (Pathwise random structured matrices). *Let $A_1, \dots, A_n$ be random matrices which almost surely lie in one of the classes covered above: scalar Toeplitz, scalar Hankel, fixed-level block Toeplitz/Hankel, a fixed finite-dimensional C^* -algebra of entropy η , or a finite group convolution algebra. On every realization for which $A_i \neq 0$, set $\tilde{A}_i = A_i / \|A_i\|_{\text{op}}$. Then the corresponding deterministic bound holds for $\tilde{A}_1, \dots, \tilde{A}_n$ on that realization.*

Remark 6.18. This pathwise statement should not be confused with deterministic Matrix Spencer for arbitrary random matrix realizations. Generic GOE, Wigner, and Wishart matrices almost surely span high-entropy subspaces of the full matrix algebra, and the finite-lift method

above does not remove the usual dimension-entropy barrier for such inputs. Average-case matrix discrepancy is a separate line of work; see, for instance, Kunisky–Zhang, Wengiel, and Maillard [18, 26, 22].

Proposition 6.19 (Entropy obstruction for exact finite block lifts). *Suppose the full algebra $M_D(\mathbb{C})$ admits an exact finite block lift through $\mathcal{B} = \bigoplus_{\alpha} M_{r_{\alpha}}(\mathbb{C})$ in the sense that the reconstruction map is onto $M_D(\mathbb{C})$. Then*

$$\eta(\mathcal{B}) = \left(\max_{\alpha} r_{\alpha} \right) \sum_{\alpha} r_{\alpha} \geq D^2.$$

Consequently, Theorem 6.3 cannot yield better than $C\sqrt{n \log(e + D^2/n)}$ for arbitrary $D \times D$ matrices through exact finite block lifts.

Proof. Surjectivity onto the D^2 -dimensional vector space $M_D(\mathbb{C})$ forces $\dim \mathcal{B} = \sum_{\alpha} r_{\alpha}^2 \geq D^2$. Since $\sum_{\alpha} r_{\alpha}^2 \leq (\max_{\alpha} r_{\alpha}) \sum_{\alpha} r_{\alpha} = \eta(\mathcal{B})$, the claim follows. $\square$

Question 6.20 (Bounded finite-entropy lift program). *Classify structured operator systems $\mathcal{S}_D \subset M_D(\mathbb{C})$ for which every contraction in $\mathcal{S}_D$ has a bounded lift either to a finite block algebra with $r_* \sum_{\alpha} r_{\alpha} = O(D)$ or to a matrix-valued multiplier model with an $O(D)$ -size norming set and bounded block size. Every such class satisfies Matrix Spencer in the regime $D \leq n$.*

The Toeplitz, Hankel, fixed-level multilevel Toeplitz, finite abelian convolution, bounded-irrep convolution, simultaneous diagonalization, bounded-block diagonalization, and commutative association-scheme examples above are all instances of this program. Solving the full Matrix Spencer conjecture will require either a way to beat the entropy obstruction for generic matrix algebras or a method that is not reducible to finite block lifts.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used generative AI tools to assist with drafting, editing, literature-check organization, and proof-audit notes. The author reviewed and edited the output, checked the mathematical claims and bibliographic information to the extent reported in the manuscript, and takes full responsibility for the content of the work.

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